

Examining Adversarial Examples Up Close

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Abstract

This project presents an interactive visualization tool for exploring adversarial examples in neural networks from a dynamical systems perspective. Building on my master's thesis research, this demonstration allows users to manipulate and visualize how different neural network architectures respond to adversarial attacks through 3D PCA analysis and customizable filtering. By providing an intuitive interface to explore the relationship between model architecture, iteration count, and adversarial robustness, this tool offers insights into whether vulnerability to such attacks is a universal property of neural networks or can be meaningfully mitigated through design choices.

1 Introduction

Adversarial examples pose significant challenges to machine learning systems, and my thesis investigated whether these vulnerabilities are universal or can be mitigated through architectural design choices. Using a dynamical systems approach, I studied how different model architectures and iterations affect robustness to targeted attacks. This visualization tool provides an intuitive interface to explore these relationships, offering insights that static visualizations cannot convey.

2 Project in a Sentence

This project creates an interactive visualization tool that allows users to explore the relationship between neural network architecture and vulnerability to adversarial examples through 3D PCA visualization and customizable filtering of experimental results.

3 Project Type & Why

This project is a compliment to my masters thesis, where I will use it as an interactive demo during my defense. The purpose of this demo is to not just regurgitate the findings I've already conveyed in my paper, but provide a cool new perspective using interactive visualizations. Additionally, I want it to be a cool interactive tool that a future student could use to learn and pick up my research where I left off and take it in new directions.

4 Audience & Needs

The audience for this project is my advisor, MS thesis committee, and any friends that come to my defense. If their needs are unmet, they will not be able to have a deeper understanding of my research. Another audience is any future student that wants to study adversarial examples in the research group. If their needs are unmet it may be more difficult to understand my work and where I've left off.

5 My Approach

My approach is to create sort of an interactive sandbox where you can see my data in all kinds of ways. The main way will be via 3D PCA analysis. However, I want to also include some neat visualizations on another page where you can read in the data but filter for different conditions. I think this approach will be successful because it builds off of things I've already created and analyzed as part of my thesis research, but expanding upon it and putting everything in one spot.

6 Best-Case Impact Statement

This interactive visualization tool successfully bridges the gap between complex technical findings and intuitive understanding, enabling researchers to visualize how different neural network architectures respond to adversarial attacks and providing compelling evidence

that while vulnerability can be meaningfully mitigated through architectural choices, the persistence of adversarial examples across diverse architectures suggests they may be a fundamental property of neural networks.

7 Major Milestones

- Design phase, create mock-ups.
- Get the shell of my interactive web app working.
- Create bins for all visualizations and map it out.
- Ensure proper data loading/handling.
- Create the PCA visualization with smaller subset of data.
- Create the visualizations with filtering and pulling data from Supabase database.

8 Anticipated Obstacles

- Navigating the size of my dataset in Supabase.
- Having enough time to complete the project as I chose to do it solo.
- Creating the 3D PCA plot as it is more complicated than doing things in 2D.

9 Relevant Publications

- **Ma, Y., Xie, T., Li, J., & Maciejewski, R. (2019). “Explaining Vulnerabilities to Adversarial Machine Learning through Visual Analytics”**

This paper presents a visual analytics framework for exploring model vulnerabilities to adversarial attacks, particularly relevant to your project as it demonstrates how interactive visualizations can effectively communicate complex relationships between models and adversarial examples, supporting your approach of using visual analytics to make your thesis findings more accessible and interactive.

- **Goodfellow, I.J., Shlens, J., & Szegedy, C. (2015). “Explaining and Harnessing Adversarial Examples”**

This seminal paper introduced gradient-based adversarial attacks and provides the foundation for the attack methods explored in your thesis.

- **Szegedy, C., et al. (2014). “Intriguing properties of neural networks”**

The first paper to formally identify adversarial examples, establishing the problem space your visualization tool explores.

- **Tanay, T., & Griffin, L. (2016). “A Boundary Tilting Perspective on the Phenomenon of Adversarial Examples”**

Offers a theoretical explanation of adversarial examples that aligns with your dynamical systems perspective.

- **Fawzi, A., et al. (2018). “Empirical Study of the Topology and Geometry of Deep Networks”**

Examines the geometric properties of decision boundaries in neural networks, which relates directly to your PCA visualization approach for understanding the spatial relationship between original images and their adversarial counterparts.

10 Measuring Success

I will know I have succeeded in this project when:

- The interactive visualization clearly demonstrates the patterns observed in my thesis, particularly the relationships between model architecture, iteration count, and adversarial robustness.
- My committee and other defense attendees can intuitively understand these complex relationships without requiring extensive technical explanations.
- The tool provides novel insights through its interactive capabilities that weren’t immediately apparent from static visualizations in my thesis.
- Future researchers can use this tool to quickly grasp the state of my research and explore areas for potential extension.
- The visualization effectively communicates both aspects of my thesis’s conclusion: that vulnerability can be mitigated through architectural choices, yet adversarial examples persist across all tested architectures, suggesting they may be a fundamental property of neural networks.